

Lecture 2 Tutorial — Answers

Answers

Q1. Key Terms

1. A parameter describes a **population**.
2. A statistic describes a **sample**.
3. A sample is a subset of a **population**.

Q2. True / False

1. **F**
2. **T**
3. **F**
4. **T**

Q3. Multiple Choice

C. Voluntary response sampling

Q4. Variable Type Classification

1. INT104 quiz score (0–20): **Quantitative, Discrete**
2. Number of AI-tool uses per week: **Quantitative, Discrete**
3. Student major (AI / CS / DS): **Categorical**
4. Time spent on revision (hours): **Quantitative, Continuous**

Q5. Missing Data

Method	One advantage	One risk
Delete cases	Simple and easy to apply	Reduces sample size and may introduce bias
Impute missing values	Keeps more data / preserves sample size	Imputed values may be inaccurate and distort results

Q6. Sampling Methods

1. **B** Systematic sampling
2. **C** Stratified sampling
3. **A** Convenience sampling
4. **D** Cluster sampling

Q7. Experimental Design

1. **Randomness:** Randomly assign students to the two revision groups.
2. **Control of confounders:** Control prior ability level.
3. **Replication:** Use many students in each group / repeat on enough participants.
4. **Outcome variable:** INT104 test score.

Q8. Histogram Concepts

1. A histogram is used to show the **distribution of a quantitative variable**.
2. If the number of bins is too small, the graph may be **too coarse and hide important patterns**.
3. If the number of bins is too large, the graph may be **too noisy and show random fluctuation**.

Q9. Distribution Shape

For a right-skewed distribution:

$$\text{Mean} > \text{Median} > \text{Mode}$$

The **right tail is longer** than the left tail.

Q10. Correlation and Significance

Given $r = 0.591$ and critical values ± 0.878 :

1. **No**
2. It is significant only if r is **greater** than $+0.878$ or **less** than -0.878 .

Q11. Boxplot Practice

Given $Q_1 = 25$, $Q_3 = 50$, and high-end outliers at 105 and 110.

- 1.

$$\text{IQR} = Q_3 - Q_1 = 50 - 25 = 25$$

2. The middle 50% of observations lie in the interval:

$$[25, 50]$$

3. **B. Right-skewed**